

# CausalMedia-GH Teacher Dashboard

How Explanation Videos Affect Student Learning in Your School

NexusLearn Group 4 | KNUST | May 2026 | Supervisor: Dr Eric Osei

## What This Report Is About

This report uses causal inference (not simple correlation) to estimate the true effect of explanation videos on student learning across 10,000 students. It controls for internet quality, prior achievement, tablet access, school resources, and 8 other background factors to isolate the actual causal contribution of videos.

## Key Findings at a Glance

| Student Group          | Video Effect (CATE) | What This Means                      |
|------------------------|---------------------|--------------------------------------|
| All students (average) | +0.0268             | Small positive effect overall        |
| Q1 — Lowest achievers  | -0.1135             | Check SHAP chart for subgroup detail |
| Q4 — Highest achievers | +0.1106             | Check SHAP chart for subgroup detail |
| Rural students         | +0.0053             | Check SHAP chart                     |
| Peri-urban students    | +0.0068             | Check SHAP chart                     |
| Urban students         | +0.0048             | Check SHAP chart                     |
| Low bandwidth (1)      | +0.0231             | Check SHAP chart                     |
| High bandwidth (3)     | -0.0211             | Check SHAP chart                     |
| No tablet access (0)   | -0.0050             | Check SHAP chart                     |
| Individual tablet (2)  | +0.0232             | Check SHAP chart                     |

Statistical note: ATE = 0.0268 (95%% CI: -0.3275 to 0.3389). CI includes zero: True. Based on 10,000 students.

## What Drives Each Student Result?

Each dot below is one student. Right side (red) = that variable pushes their video benefit up. Left side (blue) = pushes benefit down. Variables ranked top-to-bottom by overall influence.

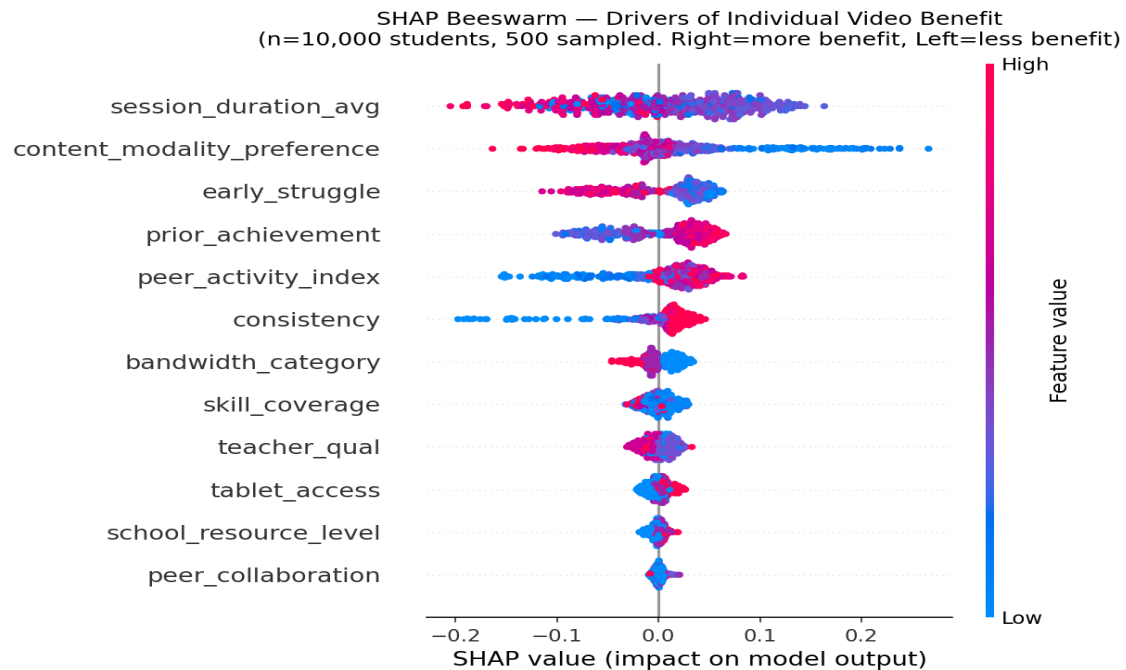


Figure 1: Global SHAP beeswarm — variable importance across all students.

## Prior Achievement and Bandwidth Interaction

This chart shows whether internet quality or prior achievement has a bigger role in determining who benefits from videos. Colour shows bandwidth level.

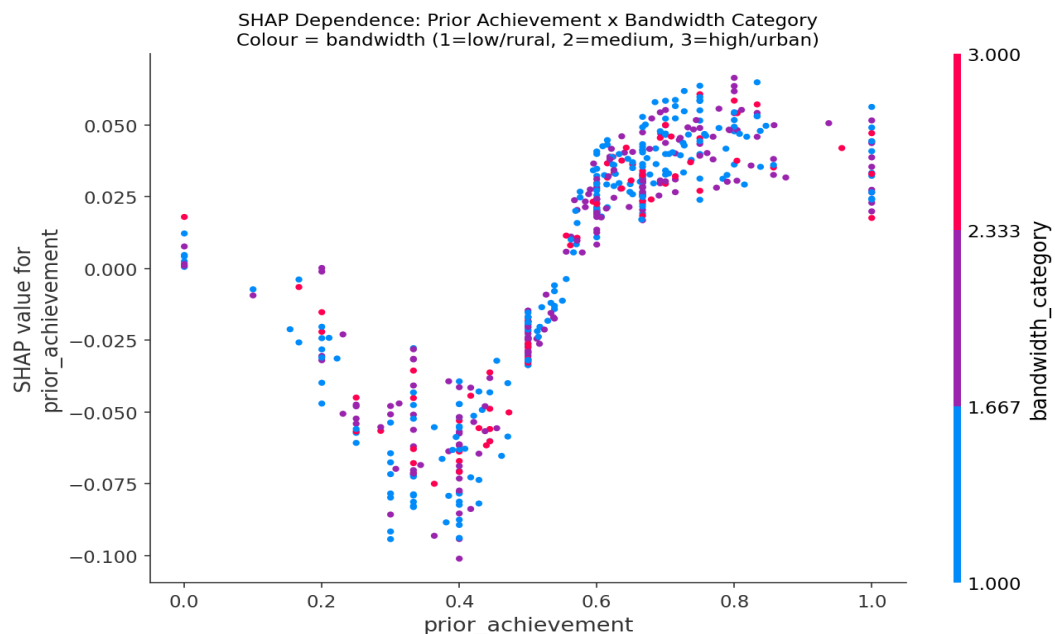
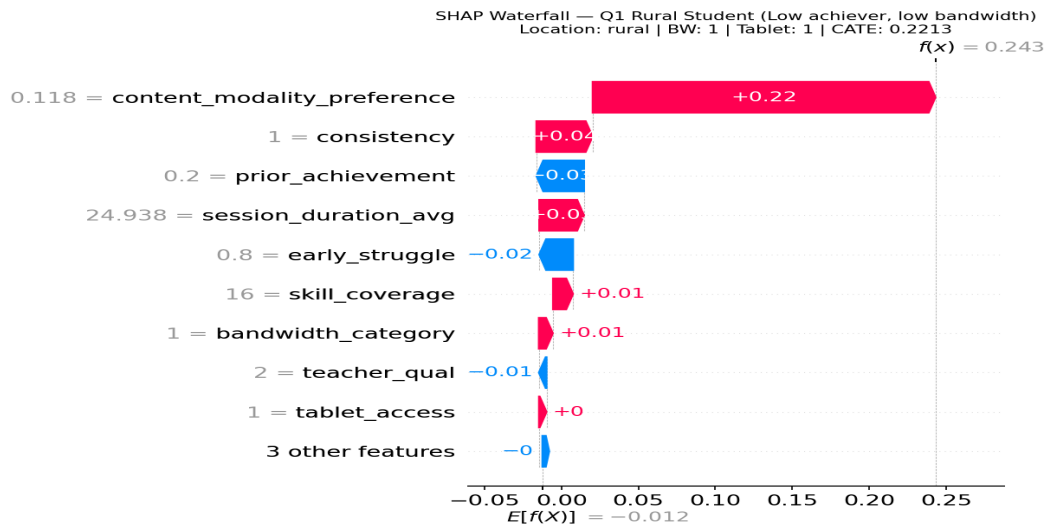


Figure 2: SHAP dependence plot — prior achievement x bandwidth.

## Three Student Profiles

### Profile A — Rural Student, Low Prior Achievement

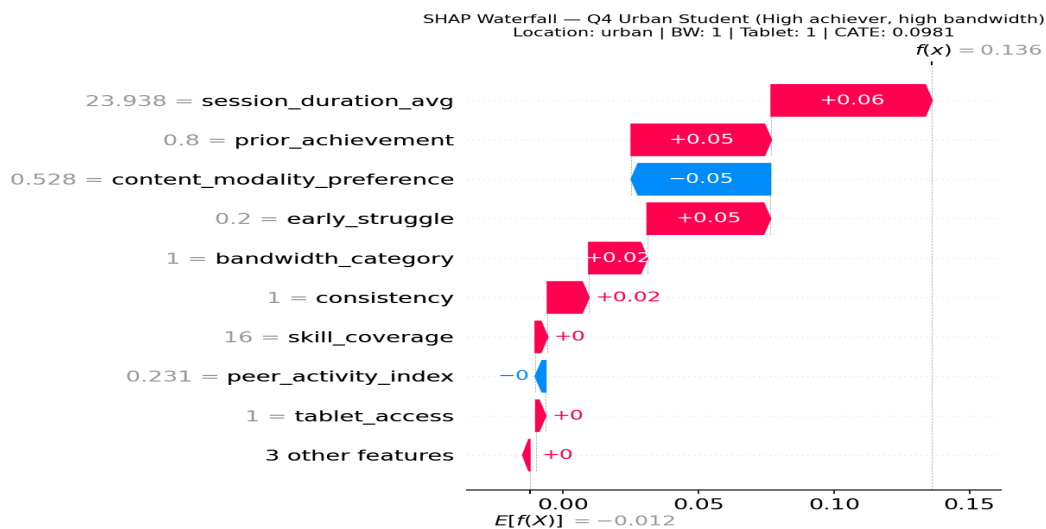
This student is in a rural area with low bandwidth and low prior achievement. The waterfall chart shows step-by-step which variables drive their individual CATE.



**Recommendation:** Prioritise video access and device support for rural low-achievers. They stand to gain the most from well-delivered multimedia content.

### Profile B — Urban Student, High Prior Achievement

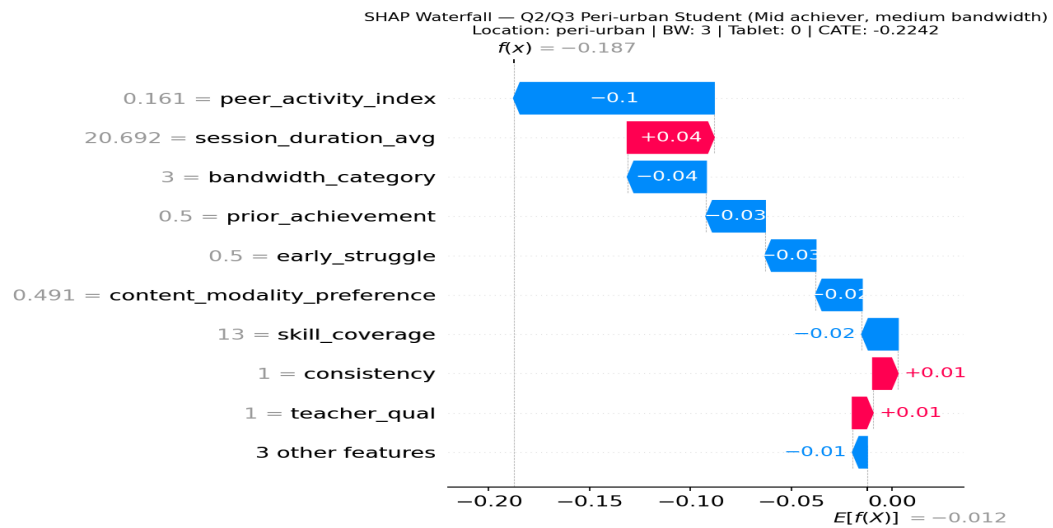
This student is in an urban school with high bandwidth and high prior achievement. The waterfall shows whether top urban students benefit or plateau from video content.



**Recommendation:** Supplement videos with harder challenge materials for top students.

### Profile C — Peri-urban Student, Mid Prior Achievement

This student is in the middle majority — peri-urban, medium bandwidth, average achievement. This group is the most heterogeneous: some benefit, some do not.



**Recommendation:** Monitor individually. Track improvement after video-heavy weeks.

## 5 Actions for Teachers

- Check the SHAP beeswarm chart for which variables matter most in your school context. The top variable is the one most worth addressing first.
- Rural and low-bandwidth students show large CATE variation. Prioritise offline video downloads so slow internet does not prevent access.
- Students with no tablet access cannot benefit from videos at all. Raise device availability with your headteacher.
- Monitor individual progress — especially for peri-urban mid-achievers where the group average hides large individual differences.
- Review this dashboard each term as new student data updates the findings.